

OPERATOR'S HANDBOOK

SATWAR

THE SHADOWING

How to play on PC, and how to win your first operation.

ALPHA 0.5 · KEYBOARD & MOUSE

WELCOME

BEFORE YOU PLAY

SATWAR runs in your browser. Open the game page, wait for it to load, and click inside it once so it gets your keyboard.

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WHAT YOU NEED

A PC
Desktop or laptop. Phones and touch screens aren't supported.

A BROWSER
Current Chrome, Edge or Firefox with WebGL 2.

MOUSE AND KEYBOARD
A wheel or trackpad for zooming.

108 MB
The first load takes a minute. Your browser then keeps it.

1920 × 1080 OR LARGER
Smaller works, but windows overlap. F11 for full screen.

THE MAIN MENU



Choose **NEW CAMPAIGN**. The game starts paused, with the tutorial on.

> **FIRST TIME?** Follow the tutorial card in the game. Each step names one thing to press and carries you to a win. This handbook is the same plan on paper.

1 · THE SITUATION

LOW JUPITER ORBIT, 2041

JOVIAN COMMAND

An unflagged inspector is creeping up on ORIEL-36, one of ours. Make it leave. No debris. No war.

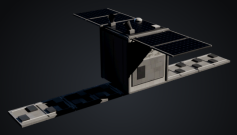
WHO'S WHO



YOU · BLUE

HYDRA BRAVO

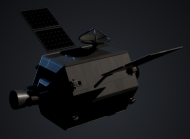
Your warship: engine, telescope, radio ears, radar and a jammer. 8.66 km/s of fuel.



PROTECT · PALE BLUE

ORIEL-36

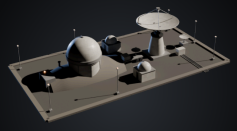
Allied radar satellite. You can't command it, and nothing may damage it.



THE INSPECTOR · YELLOW

TRK 4417

Closing from 420 km behind ORIEL-36. A distant handler radios every move. That link is its weak point.

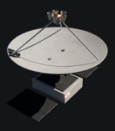


YOUR EARS · BLUE

GANYMEDE STATION

A big dish on Ganymede. Hears radio from millions of km away, but can't move.

HOW YOU WIN

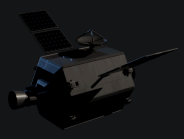


JAM ITS LINK
Press 7

→

3 h
HOLD IT
within 200 km of ORIEL-36

→



IT LEAVES
You win

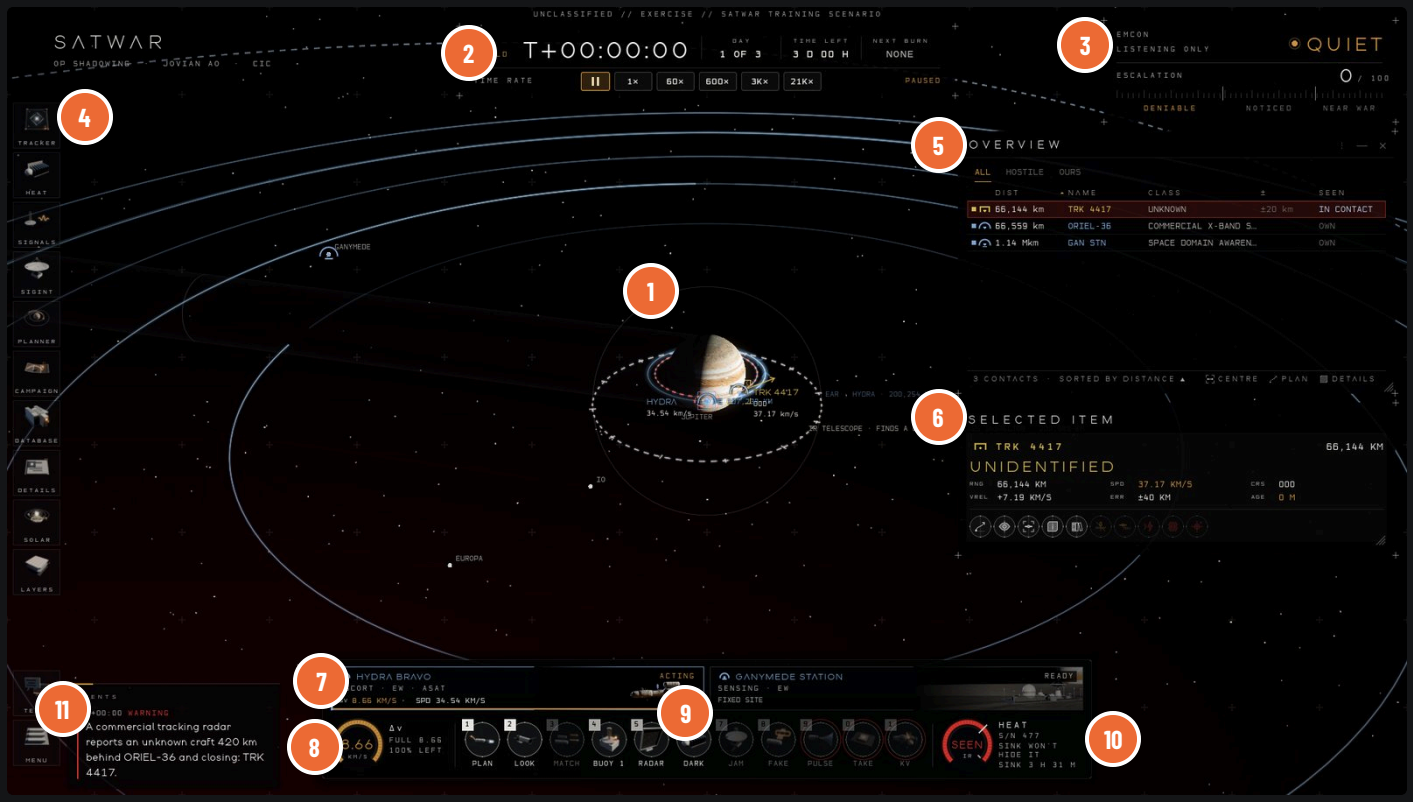
YOU LOSE IF

- **IT FINISHES** its seventh and last move
- **ANY DEBRIS** is left in orbit
- **BONUS:** identify its radio signal
- **ORIEL-36** is damaged
- **ESCALATION** reaches 100: open war

Press F1 in the game to see these goals ticked off.

ONE SCREEN, ELEVEN PARTS

Hover over anything in the game and a card explains it.



1. **MAP** Jupiter, its moons and every craft you know of. A ring round a contact shows how unsure you are of its position.
2. **CLOCK** Time since the start. Pause, or run up to 21,000×.
3. **EMCON & ESCALATION** QUIET means you only listen. Escalation 100 loses.
4. **RAIL** Opens each window: tracker, heat, SIGINT, planner and more.
5. **OVERVIEW** Every contact, nearest first. Click a row to select it.
6. **SELECTED ITEM** All you know about the contact you picked.
7. **YOUR ASSETS** Click one to act from it. HYDRA moves; the station only listens.
8. **FUEL (ΔV)** How much you can still change speed. Every burn spends it.
9. **DOCK** Your actions on keys 1–0. Grey means not now: hover to see why.
10. **HEAT** How visible you are in infrared: SEEN, WARM or COLD.
11. **EVENTS** The latest news: ADVISORY, CAUTION or WARNING.

MAP COLOURS Blue · ours · Pale blue · protected · Yellow · unknown · Red · hostile

Dashed · last known position

THE DOCK

Everything HYDRA can do, on the number keys. Quiet actions are safe; loud ones add escalation.



PLAN
Open the planner on your target

QUIET · USES FUEL



LOOK
Point the telescope at it

QUIET



MATCH
Brake alongside your target

BURNS ARE SEEN



BUOY
Drop a second listener

QUIET



RADAR
Exact range; everyone hears it

+1 PER HOUR



DARK
Hide your heat about 3.5 h, once

QUIET



JAM
Cut its orders: how you win

+2 PER HOUR



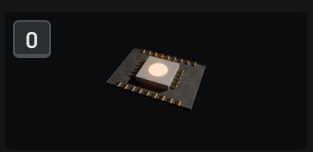
FAKE
Spoof its ranging so it drifts

+4



PULSE
Microwave burst at its receiver

+15



TAKE
Take over its controls

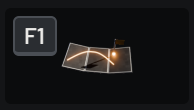
+10



KV
Kinetic kill, left without a key on purpose

+45 · DEBRIS LOSES

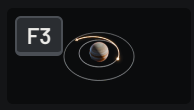
WINDOWS



CAMPAIGN
Your goals



SIGINT
Listen, locate



PLANNER
Plan a move



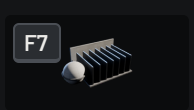
OVERVIEW
All contacts



DATABASE
Known classes



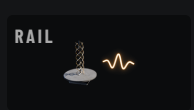
TRACKER
Track detail



HEAT
When you're seen



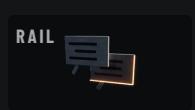
LAYERS
What's drawn



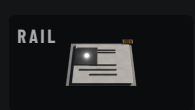
SIGNALS
All heard



SOLAR
The planets



TERMS
Plain or military



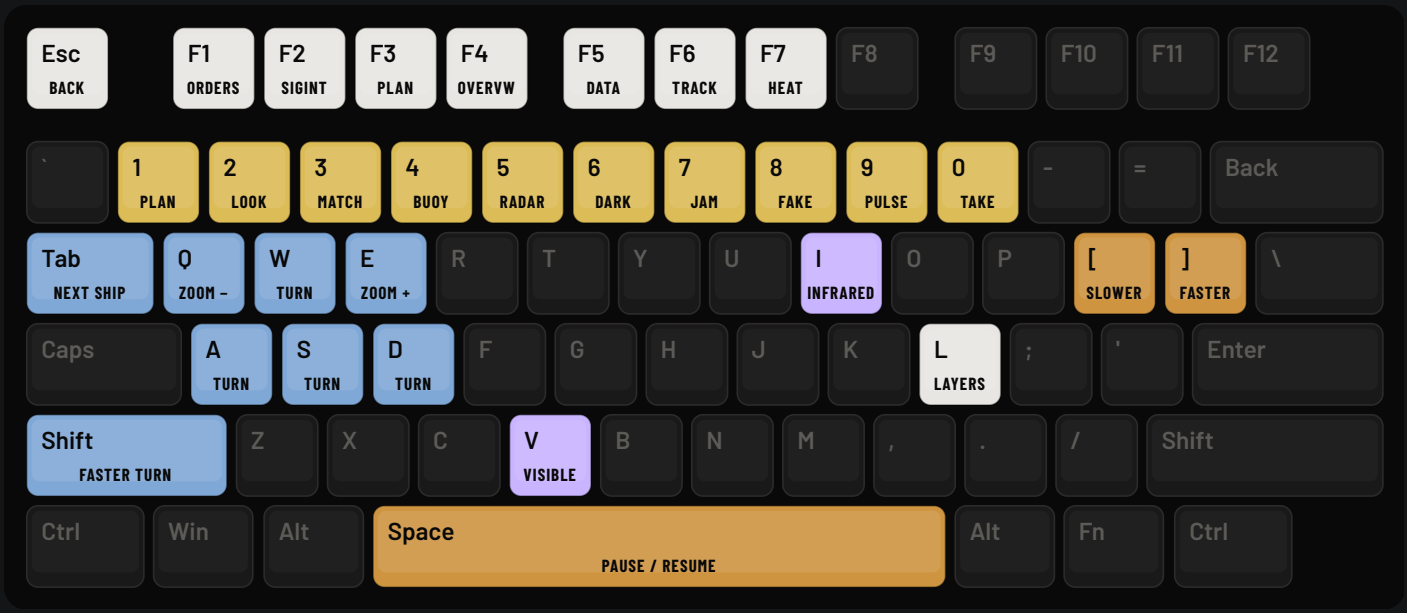
DETAILS
The maths

RAIL: no key. Click it on the left-hand rail.

4 · CONTROLS

KEYS AND MOUSE

Only the coloured keys matter. Some browsers keep F1–F7 for themselves until you have clicked inside the game.



Time · Moving the view · Windows · Dock actions · Camera



LEFT DRAG · CLICK	Turn the view · select a contact
RIGHT DRAG · CLICK	Slide the view · right-click a contact for every option and its cost
WHEEL	Zoom; keep going out to see the solar system
DOUBLE-CLICK	Centre on a ship, moon or planet and follow it

INSIDE SIGINT (F2)

↑ ↓ pick a recording · D find direction · I identify · J jam · B next band

TIME

Space pause and resume · [ ] slower and faster. Use 60× near a target, 600–3,000× while coasting, and pause to think.

PART 1: LISTEN

JOVIAN COMMAND

It reports home after every move in a short radio burst. If it sees a warship coming, it goes quiet for six hours. Listen first, from far away.



1 DROP A BUOY

PRESS `4` BUOY

A second listener where you are. Two listeners far apart can pinpoint a radio.



2 START THE CLOCK

`]` FASTER · `SPACE` PAUSE

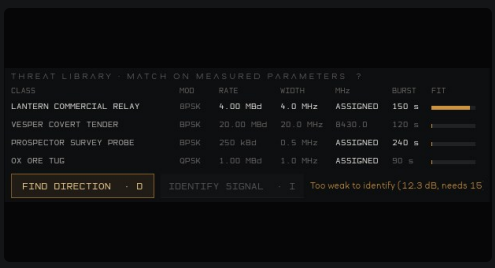
Run at 60× or more and watch TRK 4417 creep toward ORIEL-36.



3 HEAR IT

`F2` THEN CLICK ITS NEWEST BURST

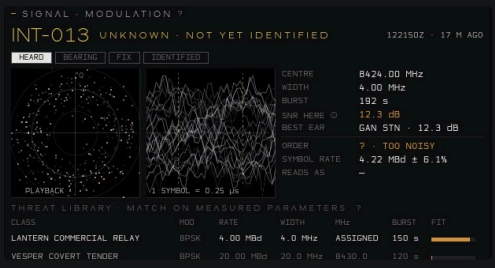
Each bright dash on the waterfall is a radio burst. Pick TRK 4417's latest.



4 FIND IT

FIND DIRECTION · `D`

Bearings from your two listeners cross where it is.



5 IDENTIFY IT

IDENTIFY SIGNAL · `I`

Names its command channel, so you can jam just that. This is the bonus goal. Too weak? Wait for a closer burst.

PART 2: CLOSE IN

JOVIAN COMMAND

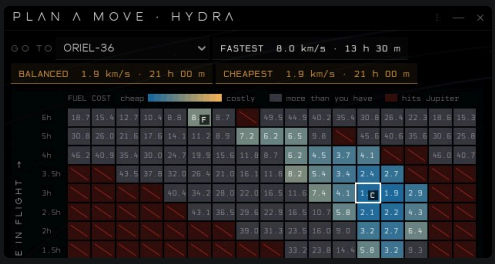
Get alongside. The planner does the orbital maths; you choose when, and how much fuel. Arriving isn't stopping: brake with MATCH.



6 PICK ORIEL-36

CLICK IT IN THE OVERVIEW

You'll meet the inspector where it's going.



7 PLAN THE TRIP

F3 · CHOOSE CHEAPEST

Each cell is a real trip: wait this long, fly this long, pay this much fuel. CHEAPEST waits 18 h, flies 3 h and costs 1.9 km/s.



8 BURN

BURN AT T+18:00

The burn is scheduled for you. Speed time up and coast.



9 MATCH

PRESS 3 MATCH ON ARRIVAL

Brakes you onto ORIEL-36's orbit, right beside it.

FUEL CHEAPEST leaves you 6.75 of your 8.66 km/s. Every extra trip spends more, and there is no refuelling.

PART 3: STAY COLD, CUT THE LINE



TRK 4417 · INTERCEPTED

MOVE COMPLETE · TARGET IN VIEW · NO ESCORT SEEN



10 GO DARK

PRESS 6 DARK · F7 FORECAST

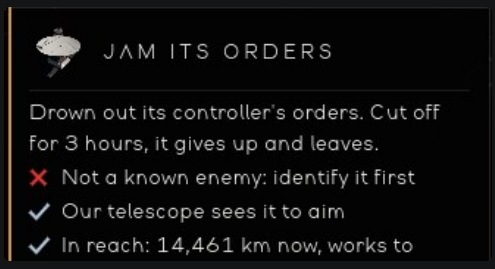
Stores your waste heat for about 3.5 h, so SEEN turns COLD. You get one, so use it once you are alongside.



11 LOOK CLOSER

SELECT TRK 4417 · 2

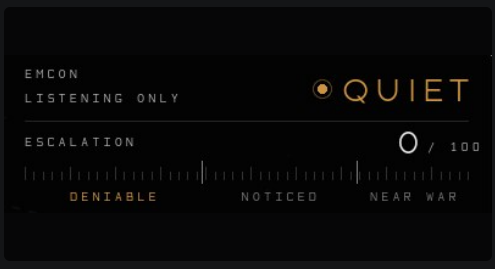
Up close the telescope sees its hull, so you can aim. V visible light, Esc closes.



12 JAM

PRESS 7 JAM

Drowns out its handler's orders. If the key is grey, hover over it: the card ticks off what's missing.



13 HOLD, THEN STOP

KEEP JAMMING 3 H · 7 AGAIN TO STOP

Cut off for three hours, it backs away and leaves. Jamming adds 2 escalation an hour, so switch it off once it's gone.

JAM IS GREY? *Not a known enemy:* identify its signal first (step 5). *Can't aim:* get closer or let the telescope see it. *Out of reach:* close in.

8 · AFTER THE OPERATION

ENDINGS

The debrief says which goal decided it and what to try next time.

THE INSPECTOR DEPARTS
It was cut off for three hours, backed away and left. No debris, no war, ORIEL-36 still working.

IT FINISHED ITS LOOK
All seven moves done. **Try:** get alongside sooner and jam earlier.

IT WENT SILENT
It saw you coming and stopped talking. **Try:** listen before you move.

OPEN WAR
Escalation reached 100. **Try:** fewer loud acts; stop jamming once it's gone.

DEBRIS
Something was destroyed in the busiest orbit there is. **Try:** never use the KV here.

FIELD NOTES



BURNS ARE SEEN
Your drive shows from millions of km. Listen before you burn.



USE THE SHADOW
F7 shows when Jupiter's shadow hides you.



A TRACK IS A BELIEF
Contacts sit where you last saw them.



QUIET BEATS LOUD
Radar and jamming cost every hour they run.



PICK THE RIGHT SPEED
60× near a target, faster while coasting.



CLICK ANY NUMBER
It opens to show how it's worked out.

HOW LOUD IS EACH ACT?



9 · CHEAT SHEET

ON ONE PAGE

Print this page and keep it beside the keyboard.

THE WINNING ORDER

1. 4 BUOY: drop a second listener
2.] speed time up to 60×
3. F2 click TRK 4417's newest burst
4. D find its direction
5. I identify its signal
6. Select ORIEL-36 · F3 CHEAPEST · BURN
7. 3 MATCH on arrival
8. 6 DARK
9. 2 LOOK at TRK 4417
10. 7 JAM and hold for 3 hours
11. 7 stop once it leaves

WIN AND LOSE

Win	Jam it 3 h within 200 km of ORIEL-36
Lose	It finishes 7 moves · ORIEL-36 damaged · any debris · escalation 100
Bonus	Identify its radio signal

KEYS

Space	Pause / resume
[ ]	Slower / faster
F1	Campaign goals
F2	SIGINT
F3	Planner
F4	Overview
F5	Database
F6	Tracker
F7	Heat forecast
L	Map layers
I V	Infrared / visible camera
Tab	Next ship
W A S D	Turn the view
Q E	Zoom
Esc	Back / menu

THE DOCK

1. PLAN
2. LOOK
3. MATCH
4. BUOY
5. RADAR · +1/h
6. DARK · once
7. JAM · +2/h
8. FAKE · +4
9. PULSE · +15
0. TAKE · +10

10 · WORDS YOU'LL SEE

GLOSSARY

TERMS on the rail switches the game between plain words and military ones.

Δv How much you can still change your speed: your fuel, in km/s.

Rendezvous Reaching another craft. Matching its orbit so you stay is the second half.

DF Direction finding. Bearings from two places cross at the transmitter.

Spot / barrage All jamming power on one known channel, or spread across a band (much weaker).

EMCON Emission control. QUIET means you only listen.

Going dark Storing waste heat instead of radiating it, so infrared loses you.

Track What you believe about a contact. Its ring shows how unsure you are.

SIGINT Signals intelligence: listening to other people's radio.

Command link The radio channel a handler uses to send a craft its orders.

J/S Jam-to-signal: how much louder you are than its handler, at its receiver.

Escalation How close the crisis is to open war. 100 is war.

RPO Rendezvous and proximity operations: one craft manoeuvring close to another.

SATWAR Operator's Handbook · Alpha 0.5. Play at kw-wargame.github.io/satwar-releases-playtest/game/. How the models work, with sources: the research edition at kw-wargame.github.io/satwar-releases-playtest/research/.

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